
Modular Inverse Reinforcement Learning via Policy Estimation and Q -Evaluation

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Inverse reinforcement learning (IRL) seeks to recover a reward function from
2 observed behavior, but the reward is typically only partially identified. We study
3 this problem in the maximum-entropy (Gumbel-shock) model under a broad class
4 of statewise affine normalization constraints that fix the mean reward under a
5 domain-informed reference policy, with anchor-action constraints as a special case.
6 This leads to *Generalized Policy-to- Q -to-Reward* (GenPQR), a modular approach
7 to normalized reward recovery based on policy estimation and Q -evaluation via
8 the Bellman equation. We establish modular finite-sample guarantees under general
9 function approximation, with terms capturing policy-estimation error and
10 Q -function estimation error. As a concrete instantiation, we study GenPQR with
11 fitted Q -evaluation, reducing IRL to policy estimation followed by regression.
12 Relative to anchor-action DeepPQR, our algorithm is simpler and more general,
13 while our analysis is broader, avoids nonstandard assumptions, and is not tied to a
14 specific neural-network architecture or training procedure.

15 1 Introduction

16 Behavioral data are abundant in robotics, economics, healthcare, and human-computer interaction.
17 Inverse reinforcement learning (IRL) seeks to explain such behavior by recovering a reward under
18 which the observed policy is optimal. Classical IRL often models agents as exactly optimal with
19 respect to an unknown reward (Ng & Russell, 2000; Abbeel & Ng, 2004), but this deterministic
20 view can miss the variability seen in real behavioral data. A common alternative is to model
21 choice stochastically, for example through entropy regularization, which yields softmax-type policies
22 (Ziebart et al., 2008, 2010; Haarnoja et al., 2017). In maximum-entropy (MaxEnt) IRL (Ziebart et al.,
23 2008), closely related to dynamic discrete-choice (DDC) models in econometrics with i.i.d. Gumbel
24 shocks (Rust, 1987), the observed policy takes a softmax form induced by an unknown reward and
25 continuation value.

26 Even in this soft setting, the reward is generally only partially identified: different reward-value
27 pairs can induce the same behavior policy through potential-based shaping transformations (Ng et al.,
28 1999; Cao et al., 2021; Skalse et al., 2023; Skalse & Abate, 2024). Unique recovery therefore requires
29 an additional normalization or structural restriction. Existing IRL methods do not by themselves
30 resolve this nonidentifiability; instead, they typically impose such restrictions through the estimation
31 procedure, for example via likelihood-based objectives, restricted reward parameterizations, or
32 adversarial formulations (Ziebart et al., 2008; Levine et al., 2011; Wulfmeier et al., 2016; Ho &
33 Ermon, 2016; Fu et al., 2018; Snoswell et al., 2020). For example, maximum-entropy IRL identifies
34 only an equivalence class of rewards unless additional restrictions, such as linear parameterizations,
35 are imposed (Ziebart et al., 2008), while adversarial IRL and related methods require stronger
36 assumptions for reward recovery, such as state-only rewards and deterministic transitions (Fu et al.,

2018). Anchor-action methods provide a useful alternative (Rust, 1987; Hotz & Miller, 1993; Aguirregabiria & Mira, 2010; Geng et al., 2020).

Building on classical DDC, Deep Policy-to- Q -to-Reward (DeepPQR) (Geng et al., 2020) shows that, under anchor-action normalization, reward recovery reduces to solving a Bellman-type fixed-point equation constructed from the observed behavior policy. This makes recovery modular: one may first estimate the behavior policy using any suitable imitation-learning or IRL method, and then estimate the required Q -function using standard value-based offline RL tools. Our work builds on and generalizes this perspective.

Our contributions. First, we characterize reward identification in maximum-entropy IRL, or equivalently in the Gumbel-shock discrete-choice model. We show that behavior alone identifies only an equivalence class of reward–value pairs, and we introduce a broad class of statewise affine normalizations that uniquely select a representative from this class, generalizing the anchor-action normalization of Geng et al. (2020). This clarifies when exact reward recovery is possible, and also shows that exact recovery is unnecessary for some policy-comparison tasks.

Second, this characterization yields a modular reward-recovery procedure. Under these normalization constraints, recovering the normalized reward reduces to two standard tasks: estimating the behavior policy and solving a linear Bellman fixed-point equation for an associated Q -function. The normalized reward–value pair is then obtained directly from this Q -function, avoiding the additional conditional-expectation step required by Geng et al. (2020). This motivates *Generalized Policy-to- Q -to-Reward* (GenPQR), which treats normalized reward recovery as a post-processing step rather than as part of a specialized joint IRL objective. As a concrete instantiation, we study GenPQR with fitted Q -evaluation (FQE). In the anchor-action special case, with a neural-network implementation, this recovers and simplifies DeepPQR.

Third, we develop a finite-sample analysis of GenPQR under general function approximation. Our analysis is modular: it can be combined with error bounds for any policy estimator and any Q -function estimator. For the FQE instantiation, the resulting bound separates the contributions of policy-estimation error, Bellman approximation error, statistical error, and iteration error. Relative to Geng et al. (2020), our theory is more general: it avoids nonstandard sup-norm assumptions on policy-estimation error, does not require Bellman completeness for FQE, and is not tied to a particular neural-network architecture or training procedure. Moreover, because GenPQR avoids the extra regression step used in Geng et al. (2020), the analysis reduces cleanly to two ingredients: a finite-sample KL bound for the policy estimator and a per-iteration regression bound.

1.1 Related work

Identifiability, shaping, and anchor-action methods. Rewards in MaxEnt IRL are only partially identified because behavior is invariant under potential-based shaping (Ng et al., 1999; Fu et al., 2018; Cao et al., 2021; Skalse et al., 2023; Skalse & Abate, 2024). Our work is most closely related to DeepPQR (Geng et al., 2020), which studies anchor-action normalization and gives finite-sample guarantees for a specific neural-network-based procedure. We extend this anchor-action view to general affine normalization constraints, clarify the identification structure, simplify the resulting recovery step (Section 3.3), and develop theory under general function approximation.

MaxEnt IRL and adversarial imitation learning. Maximum-entropy IRL and related methods fit stochastic policies induced by soft Bellman equations, often under structured reward parameterizations (Ziebart et al., 2008, 2010; Levine et al., 2011; Wulfmeier et al., 2016; Zeng et al., 2022). Adversarial approaches such as GAIL and AIRL are effective for imitation learning and behavior-policy estimation through joint reward-policy optimization (Ho & Ermon, 2016; Fu et al., 2018). As noted by Geng et al. (2020), however, these methods generally do not resolve reward nonidentifiability without stronger assumptions, such as state-only rewards. Our approach complements this literature by separating behavior-policy estimation from normalized reward recovery in a second stage, while allowing action-dependent rewards. Thus, methods designed primarily to reproduce behavior, rather than identify a unique reward, can still serve as the first stage of a modular reward-recovery procedure.

Entropy-regularized RL and control-as-inference. Our analysis is also connected to entropy-regularized control and the broader control-as-inference perspective (Kappen, 2005; Todorov, 2009; Levine, 2018), including path-consistency learning (Nachum et al., 2017), soft actor-critic (Haarnoja et al., 2018), and entropy-regularized offline RL (Haarnoja et al., 2017; Uehara et al., 2023). We

use the same soft Bellman structure, but for the inverse problem: the behavior policy is assumed to solve an entropy-regularized control problem for an unknown reward, and our goal is to recover that reward.

Value-based offline RL. The recovery step is closely connected to value-based offline RL. Fitted Q -iteration and fitted Q -evaluation estimate Bellman fixed points by regression (Ernst et al., 2005; Munos & Szepesvári, 2008; Mnih et al., 2013; van der Laan & Kallus, 2025), while minimax and critic-based methods relax completeness assumptions (Uehara et al., 2020, 2021, 2023; Xie & Jiang, 2020, 2021; Zhan et al., 2022). We do not introduce a new RL update rule; rather, we show that normalized reward recovery reduces to solving a linear fixed-point equation with these existing tools. Our finite-sample analysis then separates first-stage policy-estimation error from second-stage value-learning error.

2 Problem setup

We consider a discounted Markov decision process with state space $\mathcal{S}$, finite action space $\mathcal{A}$, transition kernel $P(\cdot \mid s, a)$, reward function $r^\dagger : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, and discount factor $\gamma \in [0, 1)$. Let $\pi(a \mid s)$ denote the behavior policy, and let ρ denote the sampling distribution over states. We observe n transitions $\{(s_i, a_i, s'_i)\}_{i=1}^n$ generated as

$$s_i \sim \rho, \quad a_i \sim \pi(\cdot \mid s_i), \quad s'_i \sim P(\cdot \mid s_i, a_i).$$

Equivalently, defining the behavior state-action distribution $\nu_\pi(ds, a) := \rho(ds)\pi(a \mid s)$, we have $(s_i, a_i) \sim \nu_\pi$ and $s'_i \sim P(\cdot \mid s_i, a_i)$. Thus, the transition kernel P is identified from the observed dynamics, whereas the reward function is not. We follow Geng et al. (2020) in focusing on finite action spaces, though the approach extends naturally to continuous actions. For any state-action function f , we write

$$(\mu f)(s) := \sum_a \mu(a \mid s) f(s, a)$$

for its expectation under policy μ at state s .

The **goal** of inverse reinforcement learning is to recover a reward function r such that π is optimal for the MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$, in an appropriate sense. We review the MaxEnt IRL setting from both the structural discrete-choice and maximum-entropy perspectives, and then formulate the optimization problem central to our analysis.

From dynamic discrete choice to MaxEnt IRL. We adopt the dynamic discrete-choice formulation (Rust, 1987; Hotz & Miller, 1993; Aguirregabiria & Mira, 2010). At time t , an agent in state s_t who takes action a_t receives utility $r^\dagger(s_t, a_t) + \varepsilon_t(a_t)$, where $r^\dagger$ is the unknown mean reward and $\varepsilon_t(a)$ is an idiosyncratic shock with known distribution. Let $V^\dagger(s)$ denote the optimal ex ante value function, define $Pf(s, a) := \mathbb{E}[f(s') \mid s, a]$, and let

$$Q^\dagger(s, a) := r^\dagger(s, a) + \gamma PV^\dagger(s, a), \quad \Xi f(s) := \log \sum_a e^{f(s, a)}.$$

Under i.i.d. type-I extreme-value shocks, the optimal policy is softmax, up to a temperature scaling,

$$\pi^\dagger(a \mid s) \propto \exp(Q^\dagger(s, a)), \quad Q^\dagger = r^\dagger + \gamma P \Xi Q^\dagger,$$

and the ex ante value satisfies $V^\dagger(s) = \Xi Q^\dagger(s)$. Equivalently, defining the state-action continuation value $v^\dagger := P \Xi Q^\dagger$, we have

$$\pi^\dagger(a \mid s) \propto \exp(r^\dagger(s, a) + \gamma v^\dagger(s, a)), \quad v^\dagger = P \Xi (r^\dagger + \gamma v^\dagger),$$

where we refer to the latter equation as the soft Bellman equation.

An equivalent perspective comes from maximum-entropy IRL (Ziebart et al., 2008, 2010), in which the agent maximizes expected discounted reward plus an entropy bonus for stochastic action selection. This yields the same optimal policy $\pi^\dagger$ and the same soft Bellman equation, where $V^\dagger$ and $Q^\dagger$ are, respectively, the entropy-regularized value and Q -functions (Haarnoja et al., 2017). Thus, despite their different motivations, the dynamic discrete-choice and MaxEnt formulations reduce to the same mathematical object: a soft Bellman system together with a softmax policy.

132 **Partial identification and the role of normalization.** Our goal is therefore to recover a reward
 133 function whose induced soft-optimal policy best matches the observed behavior, for example by
 134 minimizing the state-averaged Kullback–Leibler divergence from $\pi(\cdot | s)$ to $\pi^*(\cdot | s)$. However,
 135 matching the policy does not in general identify the reward. Under a softmax policy, adding a
 136 state-dependent offset to all action values leaves the policy unchanged, so many reward functions
 137 induce the same behavior policy (Ziebart et al., 2010; Cao et al., 2021). The reward is therefore
 138 only *partially identified*, up to an equivalence class, and selecting a unique representative requires a
 139 normalization constraint.

140 To do so, we study a broad class of **statewise affine normalization constraints**. Let $\mu(\cdot | s)$ be a
 141 reference distribution over actions, and let $g : \mathcal{S} \rightarrow \mathbb{R}$ be a specified anchor function. We impose

$$\sum_a \mu(a | s) r(s, a) = g(s) \quad \text{for all } s, \quad (1)$$

142 or equivalently, $\mu r = g$. Thus, the μ -average reward at state s is fixed at $g(s)$, with μ and g
 143 typically chosen based on domain knowledge. Our analysis also extends to more general nonlinear
 144 normalization constraints, though uniqueness may fail and the resulting recovery procedure may be
 145 less explicit.

146 Such constraints are standard in economics and are often substantively meaningful (Hotz & Miller,
 147 1993; Bajari et al., 2010). Our formulation fixes the μ -average reward under a reference policy μ ,
 148 allowing both classical and data-driven normalizations. The anchor-action constraint $r(s, a^\dagger) = g(s)$
 149 is the special case $\mu(a | s) = 1\{a = a^\dagger\}$ (Geng et al., 2020); when $g \equiv 0$, it reduces to the
 150 reference-action zero-reward normalization $r(s, a^\dagger) = 0$ (Rust, 1987). This is common in dynamic
 151 discrete-choice models, where a no-action, status-quo, or outside option is assigned zero or otherwise
 152 known payoff (Rust, 1987; Hotz & Miller, 1993; Aguirregabiria & Mira, 2010; Geng et al., 2020). For
 153 example, in Rust’s engine-replacement model, not replacing the engine has zero immediate payoff,
 154 while in marketing or uplift settings a “do nothing” action may naturally be assigned zero reward
 155 (Sawant et al., 2018). If instead $\mu(a | s) = 1/|\mathcal{A}|$, the constraint fixes the statewise average reward
 156 and, when $g \equiv 0$, reduces to a sum-to-zero constraint (Kallus & Udell, 2016). The same formulation
 157 also allows data-driven normalizations. For example, in medical decision problems one may wish to
 158 normalize rewards relative to standard care using historical outcome data (Kallus et al., 2018). If an
 159 auxiliary dataset records only states and realized outcomes (s_i, y_i) from a population with the same
 160 reward function and behavior policy π , then $g(s) = \mathbb{E}[y | s] = \sum_a \pi(a | s) r(s, a)$, so taking $\mu = \pi$
 161 yields a natural normalization.

162 **Main problem:** Putting these pieces together, we obtain the following constrained maximum-
 163 likelihood problem for recovering (r^*, v^*) :

$$\begin{aligned} \arg \max_{r, v} \quad & \mathbb{E}_{(s, a) \sim \nu_\pi} [r(s, a) + \gamma v(s, a) - \exp\{\Xi(r + \gamma v)(s)\}] \\ \text{s.t.} \quad & v = P\Xi(r + \gamma v), \quad (\text{soft Bellman}) \\ & \mu r = g. \quad (\text{affine normalization}) \end{aligned} \quad (2)$$

164 That is, we maximize the conditional log-likelihood over state–action functions r and v subject to the
 165 soft Bellman equation and a statewise affine normalization constraint.

166 The remainder of the paper shows that solving (2) reduces to two steps: estimate π , then solve a linear
 167 Bellman evaluation equation to recover the unique pair (r^*, v^*) . We then turn this characterization
 168 into a simple algorithm.

169 3 From partial to point identification

170 To expose the core structure of the problem, we first drop the normalization constraint and study the
 171 resulting relaxed optimization problem.

172 **Relaxed problem:** We remove the reward normalization in (2) but keep soft Bellman consistency:

$$\begin{aligned} \arg \max_{r, v} \quad & \mathbb{E}_{(s, a) \sim \nu_\pi} [r(s, a) + \gamma v(s, a) - \exp\{\Xi(r + \gamma v)(s)\}] \\ \text{s.t.} \quad & v = P\Xi(r + \gamma v). \quad (\text{soft Bellman}) \end{aligned} \quad (3)$$

173 This problem is highly non-unique. Because the objective is invariant under potential-based trans-
 174 formations of the reward, the relaxed problem admits an entire equivalence class of solutions. We
 175 will exploit this invariance later to recover a solution to the normalized problem (2) from a single
 176 convenient representative.

177 3.1 Behavior cloning solves the relaxed problem

178 Without normalization, the relaxed problem admits a particularly simple solution: estimate the
 179 behavior policy π and set $r(s, a) = \log \pi(a | s)$ and $v(s) = 0$. This pair maximizes the conditional
 180 log-likelihood and satisfies the soft Bellman equation. Indeed, $P\Xi(r + \gamma v) = 0$ since $\exp\{\Xi(r +$
 181 $\gamma v)(s)\} = \sum_a \exp\{r(s, a)\} = \sum_a \pi(a | s) = 1$. We will use the shorthand

$$u^*(s, a) := \log \pi(a | s).$$

182 **Lemma 1** (Trivial optimum of the relaxed problem). *The pair $(r, v) := (u^*, 0)$ solves (3).*

183 A related observation appears in Section 4 of Fu et al. (2018), but not as a tool for identification or
 184 estimation. Here, by contrast, it is the starting point for normalization-based identification. As shown
 185 in Section 3.4, this trivial solution already suffices for policy-value comparisons.

186 3.2 An invariance among solutions

187 The relaxed problem is invariant to *potential-based shaping*: adding a state-only potential $c : \mathcal{S} \rightarrow \mathbb{R}$
 188 shifts all logits in the softmax by the same amount per state, leaving both feasibility and likelihood
 189 unchanged. This is the entropy-regularized analogue of reward shaping in classical RL (Ng et al.,
 190 1999) and explains why the relaxed objective is flat along an affine subspace.

191 **Lemma 2** (Potential-based shaping invariance). *Let (r, v) be feasible for (3), and let $c : \mathcal{S} \rightarrow \mathbb{R}$
 192 be arbitrary. Define $\tilde{r} = r + c - \gamma Pc$ and $\tilde{v} = v + Pc$. Then $(\tilde{r}, \tilde{v})$ is also feasible for (3)
 193 and attains the same objective value as (r, v) . In particular, the induced log-policy $u^*(s, a) =$
 194 $r(s, a) + \gamma v(s, a) - \Xi(r + \gamma v)(s)$ is unchanged.*

195 Related partial-identification results appear in Theorem 1 of Cao et al. (2021), the discussion of
 196 reward ambiguity in Fu et al. (2018), and Lemma 2 of Geng et al. (2020). We now turn to our first
 197 main contribution: showing that an explicit normalization constraint selects a unique representative
 198 from this class.

199 3.3 Solving the original (normalized) problem

200 Lemma 1 gives one relaxed optimum, $(u^*, 0)$, and Lemma 2 characterizes all others via potential-
 201 based transformations $(u^* + c - \gamma Pc, Pc)$. Since the constrained and relaxed problems have the
 202 same optimal value, solving (2) amounts to selecting the unique relaxed optimum that satisfies the
 203 normalization $\mu r = g$, or equivalently, the unique c that enforces this constraint.

204 The next result gives this representative explicitly. Define $Q_{u^*-g}^\mu$ as the Q -function of the nor-
 205 malization policy μ under reward $u^* - g$, where $u^* = \log \pi$. Recall that $(P\mu Q)(s, a) :=$
 206 $\mathbb{E}_{s' \sim P(\cdot | s, a), a' \sim \mu(\cdot | s')} [Q(s', a')]$.

207 **Theorem 1** (IRL via a Bellman equation). *Let $Q_{u^*-g}^\mu$ be the unique bounded solution to*

$$Q_{u^*-g}^\mu(s, a) = u^*(s, a) - g(s) + \gamma(P\mu Q_{u^*-g}^\mu)(s, a).$$

208 *Then (2) admits a unique optimal solution (r^*, v^*) , given by*

$$\begin{aligned} r^*(s, a) &= Q_{u^*-g}^\mu(s, a) - (\mu Q_{u^*-g}^\mu)(s) + g(s), \\ v^*(s, a) &= \frac{1}{\gamma} (u^*(s, a) - g(s) - Q_{u^*-g}^\mu(s, a)). \end{aligned}$$

209 This theorem is the paper’s main result. It shows that normalized reward recovery reduces to
 210 estimating $u^* = \log \pi$ and then solving a linear fixed-point equation for $Q_{u^*-g}^\mu$, from which the
 211 normalized reward follows explicitly. By Lemma 2, every feasible representative in the shaping class
 212 induces the same u^* . Thus, once one feasible representative is known, a different normalization can
 213 be imposed without re-solving the original IRL problem.

214 **Comparison to DeepPQR.** In the anchor-action case $\mu(a | s) = 1(a = a^\dagger)$, our result is closely
 215 related to DeepPQR (Geng et al., 2020), but it gives a more direct characterization and also recovers
 216 the continuation value v^* . DeepPQR begins with the anchored state-value function

$$Q^{a^\dagger}(s) = g(s) + \gamma P(Q^{a^\dagger} - u^*)(s, a^\dagger).$$

217 Equivalently, $Q^{a^\dagger}$ is the value function of the deterministic policy $a \equiv a^\dagger$ under reward $r(s, a^\dagger, s') =$
 218 $g(s) - \gamma u^*(s', a^\dagger)$. It then defines $Q(s, a) = u^*(s, a) - u^*(s, a^\dagger) + Q^{a^\dagger}(s)$, and recovers the reward
 219 via

$$r^*(s, a) = Q(s, a) - \gamma P(Q(\cdot, a^\dagger) - u^*)(s, a).$$

220 Thus, DeepPQR requires an additional recovery step, implemented in practice as a separate regression
 221 of $Q(s', a^\dagger) - u^*(s', a^\dagger)$ on the current state and action. Moreover, learning $Q^{a^\dagger}$ typically uses
 222 only data with $A = a^\dagger$, so it cannot pool information across actions and may be inefficient when
 223 the anchor action is rare. By contrast, our result identifies the normalized reward directly from
 224 a single state-action Q -function, $Q_{u^*-g}^\mu$, avoiding this extra step and allowing information to be
 225 shared across actions. This object also has a standard RL interpretation: $Q_{u^*-g}^\mu$ is the Q -function
 226 of the normalization policy μ under reward $u^* - g$, and the normalized reward is the corresponding
 227 advantage function, $Q_{u^*-g}^\mu(s, a) - (\mu Q_{u^*-g}^\mu)(s)$, shifted by $g(s)$.

228 3.4 Behavior cloning suffices for policy comparison

229 For policy comparison, exact reward recovery is unnecessary once the transition kernel P and discount
 230 factor γ are fixed. Let $V_r^\pi = \pi Q_r^\pi$ denote the value function under reward r and policy π , where Q_r^π
 231 is the corresponding Q -function.

232 **Theorem 2** (Identification of policy value differences). *Let (r, v) solve (3), for example $(u^*, 0)$.
 233 Then, for any policy π , $Q_{r^*}^\pi(s, a) = Q_r^\pi(s, a) - v(s, a)$. Consequently, for any two policies π_1, π_2 ,
 234 $V_{r^*}^{\pi_1}(s) - V_{r^*}^{\pi_2}(s) = V_r^{\pi_1}(s) - V_r^{\pi_2}(s)$.*

235 Thus, exact reward recovery is needed only for targets that depend on the normalization itself,
 236 such as evaluation under counterfactual transition dynamics or discount factors, or interpretation of
 237 structural features of r . Otherwise, u^* , or any reward solving the relaxed problem, suffices for policy
 238 comparison. This parallels Hotz & Miller (1993), where under Gumbel shocks value differences are
 239 identified from log-odds of observed choices.

240 4 A generic algorithm

241 Algorithm 1 presents Generalized Policy-to- Q -to-Reward (GenPQR) for reward recovery. The
 242 method has two standard steps. First, we estimate the behavior policy $\pi(a | s)$, or equivalently its
 243 log-policy $u^*(s, a) = \log \pi(a | s)$, yielding an estimate $\hat{u}$. Second, we recover (r^*, v^*) by estimating
 244 the Q -function of policy μ under reward $\hat{u} - g$, and then applying the plug-in formula from Theorem 2.
 245 When $\hat{\pi}(a | s)$ takes values near 0, $\hat{u}$ may be unstable; in practice, one may clip $\hat{\pi}$ away from zero,
 246 for example by imposing a lower bound on the estimated policy probabilities (Ionides, 2008). Thus,
 247 in our setting, IRL reduces to behavior-policy estimation plus offline policy evaluation, rather than a
 248 fundamentally new estimation problem.

249 The procedure is modular and black-box. The first stage can use any probabilistic classifier, behavior-
 250 cloning, imitation-learning, or IRL method trained on the observed data, including MaxEnt IRL
 251 and adversarial IRL. The second stage can use a range of existing offline RL methods, including
 252 temporal-difference learning (Tsitsiklis & Van Roy, 1996), fitted Q -evaluation (FQE) (Munos &
 253 Szepesvári, 2008; van der Laan & Kallus, 2025), and minimax or saddle-point Q -learning methods
 254 (Uehara et al., 2020, 2021; Xie & Jiang, 2020, 2021). This separation lets each stage leverage existing
 255 methods and remain compatible with techniques for handling misspecification or distribution shift
 256 (Fujimoto et al., 2019; Agarwal et al., 2021; Chen & Jiang, 2019; Foster et al., 2021). Moreover, given
 257 any feasible reward-value pair under one normalization, GenPQR can recover the corresponding
 258 reward under another normalization without re-solving the original IRL problem.

259 Algorithm 2 instantiates the second stage of GenPQR using FQE, thereby generalizing DeepPQR
 260 (Geng et al., 2020). It requires only standard tools for policy estimation and regression. Moreover,

Algorithm 1 GENERALIZED POLICY-TO-Q-TO-REWARD (GENPQR)

input Transitions $\{(s_i, a_i, s'_i)\}_{i=1}^n$, normalization $\mu(a | s)$, anchor $g(s)$, discount γ

- 1: **Policy estimation:** estimate $\hat{u}(s, a) \approx \log \pi(a | s)$ using, for example, classification, behavior cloning, or another IRL method
- 2: **Q-evaluation:** solve

$$\hat{Q}(s, a) \approx \hat{u}(s, a) - g(s) + \gamma(P\mu\hat{Q})(s, a)$$
 using any approximate dynamic-programming or policy-evaluation method

output Reward:

$$\hat{r}(s, a) = \hat{Q}(s, a) - \sum_{a'} \mu(a' | s) \hat{Q}(s, a') + g(s)$$

Algorithm 2 FITTED Q EVALUATION for $Q_{\hat{u}-g}^\mu$

input Transitions $\{(s_i, a_i, s'_i)\}_{i=1}^n$, log-policy $\hat{u}$, normalization $\mu(a | s)$, anchor $g(s)$, discount γ , class $\mathcal{F}$, iterations K

- 1: Initialize $\hat{Q}^{(0)}(s, a) \leftarrow 0$
- 2: **for** $k = 1, \dots, K$ **do**
- 3: For each i , set

$$y_i \leftarrow \hat{u}(s_i, a_i) - g(s_i) + \gamma \sum_{a'} \mu(a' | s'_i) \hat{Q}^{(k-1)}(s'_i, a')$$
- 4: Fit $\hat{Q}^{(k)} \in \mathcal{F}$ by regressing y_i on (s_i, a_i)
- 5: **end for**

output $\hat{Q} \leftarrow \hat{Q}^{(K)}$

the regression step in Algorithm 2 need not be solved exactly at each iteration: one may instead parametrize Q by a neural network and take one or a few stochastic-gradient steps per iteration, as in deep fitted Q -learning or boosting (Tosatto et al., 2017). The expectation under μ can also be approximated by Monte Carlo sampling to avoid explicit integration.

5 Theoretical Guarantees

5.1 A generic and modular deterministic bound

We now turn to finite-sample guarantees for $\hat{r}(s, a) := \hat{Q}(s, a) - (\mu\hat{Q})(s) + g(s)$. The following result shows how errors in the estimated log-policy $\hat{u}$ and anchored Q -function $\hat{Q}$ propagate to the recovered reward.

We measure errors in the behavior norm, but the Bellman operator is most naturally analyzed in $L^2(d_\mu)$, where d_μ is any stationary state distribution under (μ, P) (Patterson et al., 2022; van der Laan & Kallus, 2025), i.e.,

$$d_\mu = d_\mu P_\mu, \quad P_\mu(s' | s) := \int_{\mathcal{A}} P(s' | s, a) \mu(a | s) da.$$

Our analysis therefore requires relating the behavior norm to the $L^2(d_\mu)$ norm. Let $\nu_\pi := \rho \otimes \pi$ denote the behavior distribution on state-action pairs, and write

$$\|f\|_{2, \text{beh}} := \left(\mathbb{E}_{(s, a) \sim \nu_\pi} [f(s, a)^2] \right)^{1/2}, \quad \|f\|_\infty := \operatorname{ess\,sup}_{s, a} |f(s, a)|$$

Informally, the following assumption requires the observed system (ρ, π, P) to provide sufficient coverage relative to the normalization policy μ under the same dynamics P : π must cover μ , the behavior state distribution ρ must cover the stationary distribution d_μ , and the stationary distribution d_μ must cover the one-step state distribution $\nu_\pi P$.

Assumption 1 (Policy mismatch, stationary-state coverage, and one-step state coverage). *Assume*

$$C_{\mu/\pi} := \sup_{s, a} \frac{\mu(a | s)}{\pi(a | s)} < \infty, \quad C_{d_\mu/\rho} := \left\| \frac{dd_\mu}{d\rho} \right\|_\infty < \infty, \quad C_{\nu_\pi P/d_\mu} := \left\| \frac{d(\nu_\pi P)}{dd_\mu} \right\|_\infty < \infty.$$

Condition 1 plays a role analogous to classical concentrability or coverage conditions in analyses of FQE and offline RL (Xie et al., 2022; Zhan et al., 2022). Because the normalization policy μ is typically chosen using knowledge of the agent or of π itself, Assumption 1 is often less restrictive in our IRL setting than in standard offline policy evaluation. For example, in the anchor-action case, it would not make sense to anchor on an action the agent never takes.

285 **Theorem 3** (Reward recovery bound). *Under Assumption 1,*

$$\|r^* - \hat{r}\|_{2,\text{beh}} \leq (1 + \sqrt{C_{\mu/\pi}}) \left\{ \|Q_{\hat{u}-g}^{\mu} - \hat{Q}\|_{2,\text{beh}} + \left(1 + \frac{\gamma \sqrt{C_{\nu_{\pi}P/d_{\mu}} C_{d_{\mu}/\rho}}}{1 - \gamma}\right) \|u^* - \hat{u}\|_{2,\text{beh}} \right\}.$$

286 The reward-estimation error decomposes into a Q -estimation term, $\|Q_{\hat{u}-g}^{\mu} - \hat{Q}\|_{2,\text{beh}}$, and a policy-
 287 estimation term, $\|\hat{u} - u^*\|_{2,\text{beh}}$. This makes the bound modular: any guarantees for Q -learning
 288 and policy estimation translate directly into a reward-recovery guarantee, with the latter scaled by
 289 $(1 - \gamma)^{-1}$ and the coverage coefficients in Assumption 1. Under sample splitting (Foster & Syrgkanis,
 290 2023), for example, existing results apply directly to $\hat{u}$ and $\hat{Q}$, including bounds for FQE (Munos &
 291 Szepesvári, 2008; van der Laan & Kallus, 2025) and minimax Q -learning (Uehara et al., 2020). In
 292 the next section, we apply this template to derive finite-sample bounds for PQR with FQE.

293 **Comparison to DeepPQR.** Theorem 2 of Geng et al. (2020) gives a related finite-sample bound
 294 for anchor-action reward recovery via FQE under a specific neural-network architecture. Beyond
 295 its limited function-approximation scope, a key restriction is its reliance on sup-norm guarantees
 296 for the learned policy, which are nonstandard for classification and may be brittle under limited
 297 coverage. By contrast, likelihood-based methods such as multiclass classification, MaxEnt IRL
 298 (Ziebart et al., 2008), and adversarial IRL (Fu et al., 2018) are more naturally analyzed in L^2 or
 299 Kullback–Leibler divergence (Snoswell et al., 2020; Ke et al., 2020; Foster et al., 2024). In Geng
 300 et al. (2020), this sup-norm requirement appears to arise from using sup-norm contraction of the
 301 Bellman evaluation operator, whereas our analysis works in a stationary L^2 norm. This yields more
 302 standard policy-estimation assumptions and makes the role of coverage in reward recovery explicit.

303 5.2 Generalization conditions for GenPQR with FQE

304 We now specialize Theorem 3 to the case in which $\hat{Q}$ is obtained by running K steps of FQE over a
 305 regression class $\mathcal{F}$, initialized at $\hat{Q}^{(0)}$, as in Algorithm 2. To state the resulting finite-sample bound,
 306 we assume PAC-style generalization guarantees for the policy-estimation and regression steps.

307 **Assumption 2** (Policy generalization). *For all $\delta \in (0, 0.5)$, suppose $\|\hat{u}\|_{\infty} \vee \|u^*\|_{\infty} \leq B < \infty$, and
 308 with probability at least $1 - \delta$, $\{\mathbb{E}_{s \sim \rho} [\text{KL}(\pi(\cdot | s) \| \hat{\pi}(\cdot | s))]\}^{1/2} \leq \hat{\rho}_{\pi}(n, \delta)$.*

309 Boundedness of $\hat{u}$ and u^* ensures that KL divergence controls $\|\hat{u} - u^*\|_{2,\text{beh}}$. For maximum likelihood
 310 estimation over a class $\mathcal{U}$, one typically has $\hat{\rho}_{\pi}(n, \delta) \lesssim r_{\mathcal{U}}(n) + \sqrt{\log(1/\delta)/n}$, where $r_{\mathcal{U}}(n)$ is a
 311 local complexity measure of $\mathcal{U}$ (e.g., VC dimension) (van de Geer, 2000; Wainwright, 2019).

312 We require each regression step to achieve small excess risk relative to the best approximation in $\mathcal{F}$
 313 to the Bellman target $\mathcal{T}_{\hat{u}}^{\mu}(\hat{Q}^{(k-1)})$, where $\mathcal{T}_{\hat{u}}^{\mu} f := u - g + \gamma P \mu f$. Define

$$\text{reg}(Q | u, Q') := \|\mathcal{T}_{\hat{u}}^{\mu}(Q') - Q\|_{2,\text{beh}}^2 - \inf_{f \in \mathcal{F}} \|\mathcal{T}_{\hat{u}}^{\mu}(Q') - f\|_{2,\text{beh}}^2.$$

314 **Assumption 3** (One-step regression generalization). *For all $k \in [K]$ and $\delta \in (0, 0.5)$, with probab-
 315 ility at least $1 - \delta$, $\{\text{reg}(\hat{Q}^{(k)} | \hat{u}, \hat{Q}^{(k-1)})\}^{1/2} \leq \hat{\rho}_Q(n, \delta)$.*

316 Such bounds for nuisance-dependent regression targets follow from Foster & Syrgkanis (2023);
 317 van der Laan (2026). Under a fresh-sample analysis (Munos & Szepesvári, 2008), one typically has
 318 $\hat{\rho}_Q(n, \delta) \lesssim r_{\mathcal{F}}(n/K) + \sqrt{\log(1/\delta)/n}$, where $r_{\mathcal{F}}(n/K)$ is the local Rademacher critical radius of
 319 $\mathcal{F}$; see Appendix C. Similar bounds also hold for weakly dependent trajectory data under suitable
 320 mixing conditions (Yu, 1994; Mohri & Rostamizadeh, 2010).

321 5.3 Finite-sample bound for GenPQR with FQE

322 We now state our main result for GenPQR with FQE under approximate Bellman-completeness. To
 323 account for misspecification in the regression steps, we introduce the *inherent Bellman error* of the
 324 regression class (Munos & Szepesvári, 2008):

$$\varepsilon_{\mathcal{F}} := \sup_{f \in \mathcal{F}} \inf_{g \in \mathcal{F}} \|\mathcal{T}_{u^*}^{\mu} f - g\|_{\infty}.$$

325 This quantity is zero when $\mathcal{F}$ is Bellman complete, that is, when $\mathcal{T}_{u^*}^\mu f \in \mathcal{F}$ for every $f \in \mathcal{F}$ (Chen
 326 & Jiang, 2019). In that case, each regression step in Algorithm 2 is correctly specified, since its
 327 population target lies in $\mathcal{F}$.

328 **Theorem 4** (Finite-sample bound for GenPQR with FQE). *Assume Assumptions 1, 2, and 3. Then,*
 329 *there exists a constant $C = C(B) \in (0, \infty)$, such that with probability at least $1 - \delta$,*

$$\begin{aligned} \|r^* - \hat{r}\|_{2,\text{beh}} \leq & (1 + \sqrt{C_{\mu/\pi}}) \left\{ \gamma^K \|\hat{Q}^{(0)} - Q_{\hat{u}-g}^\mu\|_\infty + \frac{1}{1-\gamma} \left(\varepsilon_{\mathcal{F}} + \hat{\rho}_Q(n, \delta/(2K)) \right) \right. \\ & \left. + \left(C + \frac{\gamma \sqrt{C_{\nu_\pi P/d_\mu} C_{d_\mu/\rho}}}{1-\gamma} \right) \hat{\rho}_\pi(n, \delta/2) \right\}. \end{aligned}$$

330 **Proof sketch.** Apply Theorem 3 and bound $\|Q_{\hat{u}-g}^\mu - \hat{Q}\|_{2,\text{beh}}$ using the inexact Picard argument
 331 of Munos & Szepesvári (2008). The main difference is that the inherent Bellman error is measured
 332 with respect to $\mathcal{T}_{u^*}^\mu$, rather than the data-dependent operator $\mathcal{T}_{\hat{u}}^\mu$. By Lemma 4, controlling the former
 333 suffices up to policy-estimation error. The approximate Bellman-completeness assumption can also
 334 be relaxed using suitable weighting (van der Laan & Kallus, 2025) or a minimax formulation (Uehara
 335 et al., 2023).

336 **Discussion.** The bound matches the usual FQE structure (Munos & Szepesvári, 2008; van der
 337 Laan & Kallus, 2025), with an additional term for estimating u^* . The coverage constants play a
 338 role analogous to concentrability coefficients. The three terms respectively capture finite-iteration
 339 error, approximation and statistical error from the fitted Bellman updates, and the IRL-specific
 340 error from estimating u^* . If $\hat{u}$ and $\hat{Q}$ are learned by ERM over parametric classes $\mathcal{U}$ and $\mathcal{F}$ with
 341 pseudo-dimensions $d_{\mathcal{U}}$ and $d_{\mathcal{F}}$, and $K \asymp \log n$, then up to coverage constants, one can typically
 342 show

$$\|\hat{r} - r^*\|_{2,\lambda_\mu} \lesssim \frac{1}{1-\gamma} \left(\varepsilon_{\mathcal{F}} + \sqrt{\frac{(d_{\mathcal{U}} \vee d_{\mathcal{F}}) + \log(\log n/\delta)}{n}} \right).$$

343 **Comparison to DeepPQR.** Compared with Theorem 2 of Geng et al. (2020) in the anchor-action
 344 setting, our result allows policy error to be controlled in Kullback–Leibler divergence rather than sup
 345 norm, permits approximate Bellman completeness with respect to $\mathcal{T}_{u^*}^\mu$, and is not tied to a specific
 346 neural-network architecture.

347 6 Experiments

A An equivalent identification

The following theorem gives an equivalent identification to Theorem 1: it first characterizes v^* as the solution to a fixed-point equation, and then recovers the reward in closed form.

Theorem 5 (An equivalent identification). *Equation (2) admits a unique optimal solution (r^*, v^*) , where*

$$r^* = u^* + \mu(\gamma v^* - u^*) - \gamma v^*,$$

and v^* is the unique bounded solution to the fixed-point equation

$$v^* = P\mu(\gamma v^* - u^*).$$

B Proof of Theorem 3

Proof of Theorem 3. For any measure d on $\mathcal{S}$, define

$$\|h\|_{2,d} := \left(\mathbb{E}_{s \sim d}[h(s)^2] \right)^{1/2}.$$

Since

$$r^*(s, a) = Q_{u^*-g}^\mu(s, a) - (\mu Q_{u^*-g}^\mu)(s) + g(s),$$

we have

$$r^* - \hat{r} = \tilde{\Delta}_Q - \mu \tilde{\Delta}_Q, \quad \tilde{\Delta}_Q := Q_{u^*-g}^\mu - \hat{Q}.$$

Thus

$$\|r^* - \hat{r}\|_{2,\text{beh}} \leq \|\tilde{\Delta}_Q\|_{2,\text{beh}} + \|\mu \tilde{\Delta}_Q\|_{2,\text{beh}}. \quad (4)$$

For any measurable $f : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, Jensen's inequality gives

$$|\mu f(s)|^2 = \left| \sum_a \mu(a | s) f(s, a) \right|^2 \leq \sum_a \mu(a | s) f(s, a)^2.$$

Therefore,

$$\begin{aligned} \|\mu f\|_{2,\text{beh}}^2 &= \int |\mu f(s)|^2 d\rho(s) \\ &\leq \int \sum_a \mu(a | s) f(s, a)^2 d\rho(s) \\ &= \int \frac{\mu(a | s)}{\pi(a | s)} f(s, a)^2 d\nu_\pi(s, a) \\ &\leq C_{\mu/\pi} \|f\|_{2,\text{beh}}^2. \end{aligned}$$

Hence

$$\|\mu f\|_{2,\text{beh}} \leq \sqrt{C_{\mu/\pi}} \|f\|_{2,\text{beh}}.$$

Applying this in (4) yields

$$\|r^* - \hat{r}\|_{2,\text{beh}} \leq (1 + \sqrt{C_{\mu/\pi}}) \|\tilde{\Delta}_Q\|_{2,\text{beh}}. \quad (5)$$

Next, write

$$\Delta Q := Q_{u^*-g}^\mu - Q_{\hat{u}-g}^\mu, \quad \Delta u := u^* - \hat{u}.$$

Then

$$\|\tilde{\Delta}_Q\|_{2,\text{beh}} \leq \|Q_{u^*-g}^\mu - \hat{Q}\|_{2,\text{beh}} + \|\Delta Q\|_{2,\text{beh}}. \quad (6)$$

Subtracting the Bellman equations gives

$$\Delta Q = \Delta u + \gamma P\mu \Delta Q,$$

and hence

$$\|\Delta Q\|_{2,\text{beh}} \leq \|\Delta u\|_{2,\text{beh}} + \gamma \|P\mu \Delta Q\|_{2,\text{beh}}.$$

367 Define the stationary state-action distribution

$$\lambda_\mu(ds, da) := d_\mu(ds)\mu(da | s).$$

368 By Jensen's inequality,

$$(P\mu\Delta Q)^2 \leq P\mu((\Delta Q)^2),$$

369 so

$$\begin{aligned} \|P\mu\Delta Q\|_{2,\text{beh}}^2 &\leq \int P\mu((\Delta Q)^2) d\nu_\pi = \int (\Delta Q)^2 d(\nu_\pi P\mu) \\ &= \int \sum_a (\Delta Q(s, a))^2 \mu(a | s) d(\nu_\pi P)(s) \\ &\leq C_{\nu_\pi P/d_\mu} \int \sum_a (\Delta Q(s, a))^2 \mu(a | s) dd_\mu(s) \\ &= C_{\nu_\pi P/d_\mu} \|\Delta Q\|_{2,\lambda_\mu}^2. \end{aligned}$$

370 It follows that

$$\|\Delta Q\|_{2,\text{beh}} \leq \|\Delta u\|_{2,\text{beh}} + \gamma \sqrt{C_{\nu_\pi P/d_\mu}} \|\Delta Q\|_{2,\lambda_\mu}. \quad (7)$$

371 It remains to bound $\|\Delta Q\|_{2,\lambda_\mu}$. Since Δu is a state function, we identify it with the state-action
372 function $(s, a) \mapsto \Delta u(s)$. Taking $L^2(\lambda_\mu)$ norms in

$$\Delta Q = \Delta u + \gamma P\mu\Delta Q$$

373 gives

$$\|\Delta Q\|_{2,\lambda_\mu} \leq \|\Delta u\|_{2,\lambda_\mu} + \gamma \|P\mu\Delta Q\|_{2,\lambda_\mu}.$$

374 Because λ_μ is stationary for $P\mu$, Jensen's inequality implies

$$\|P\mu f\|_{2,\lambda_\mu} \leq \|f\|_{2,\lambda_\mu} \quad \text{for all } f.$$

375 Thus

$$(1 - \gamma) \|\Delta Q\|_{2,\lambda_\mu} \leq \|\Delta u\|_{2,\lambda_\mu}.$$

376 Since Δu does not depend on a ,

$$\|\Delta u\|_{2,\lambda_\mu}^2 = \int (\Delta u(s))^2 d_\mu(s).$$

377 By the state coverage assumption,

$$\|\Delta u\|_{2,\lambda_\mu}^2 = \int (\Delta u(s))^2 d_\mu(s) \leq C_{d_\mu/\rho} \int (\Delta u(s))^2 d\rho(s) = C_{d_\mu/\rho} \|\Delta u\|_{2,\text{beh}}^2.$$

378 Hence

$$\|\Delta Q\|_{2,\lambda_\mu} \leq \frac{\sqrt{C_{d_\mu/\rho}}}{1 - \gamma} \|u^* - \hat{u}\|_{2,\text{beh}}. \quad (8)$$

379 Combining (6), (7), and (8) yields

$$\|\tilde{\Delta}_Q\|_{2,\text{beh}} \leq \|Q_{\hat{u}-g}^\mu - \hat{Q}\|_{2,\text{beh}} + \left(1 + \frac{\gamma \sqrt{C_{\nu_\pi P/d_\mu} C_{d_\mu/\rho}}}{1 - \gamma}\right) \|u^* - \hat{u}\|_{2,\text{beh}}.$$

380 Substituting this bound into (5) proves the result. $\square$

381 C Technical lemmas

382 **Lemma 3.** *Under Assumption 2, there exists a constant $C \in (0, \infty)$ such that, with probability at*
383 *least $1 - \delta$,*

$$\|\hat{u} - u^*\|_{2,\text{beh}} \leq C \hat{\rho}_\pi(n, \delta).$$

384 *Proof.* Let $u^* = \log \pi$, $\hat{u} = \log \hat{\pi}$, and define the likelihood ratio

$$\vartheta(a | s) := \frac{\hat{\pi}(a | s)}{\pi(a | s)}.$$

385 Since u^* and $\hat{u}$ are uniformly bounded, there exists $L < \infty$ such that

$$|\log \vartheta(a | s)| = |\hat{u}(s, a) - u^*(s, a)| \leq L$$

386 for all (s, a) . Equivalently, $\vartheta(a | s) \in [e^{-L}, e^L]$.

387 Now define $\phi(t) := -\log t - (1 - t)$. Since $\phi''(t) = t^{-2}$, the function ϕ is e^{-2L} -strongly convex
388 on $[e^{-L}, e^L]$. Also, $\mathbb{E}_{\pi(\cdot | s)}[\vartheta] = 1$, so for each s ,

$$\begin{aligned} \text{KL}(\pi(\cdot | s) \parallel \hat{\pi}(\cdot | s)) &= \mathbb{E}_{\pi(\cdot | s)}[-\log \vartheta] = \mathbb{E}_{\pi(\cdot | s)}[\phi(\vartheta)] \\ &\geq \frac{e^{-2L}}{2} \mathbb{E}_{\pi(\cdot | s)}[(\vartheta - 1)^2]. \end{aligned}$$

389 Moreover, by the mean value theorem, for $t \in [e^{-L}, e^L]$,

$$|\log t| \leq e^L |t - 1|.$$

390 Applying this with $t = \vartheta$ gives

$$\begin{aligned} \mathbb{E}_{\pi(\cdot | s)}[(\hat{u} - u^*)^2] &= \mathbb{E}_{\pi(\cdot | s)}[(\log \vartheta)^2] \\ &\leq e^{2L} \mathbb{E}_{\pi(\cdot | s)}[(\vartheta - 1)^2] \\ &\leq 2e^{4L} \text{KL}(\pi(\cdot | s) \parallel \hat{\pi}(\cdot | s)). \end{aligned}$$

391 Averaging over s yields

$$\begin{aligned} \|\hat{u} - u^*\|_{2, \text{beh}} &\leq \sqrt{2} e^{2L} \left\{ \mathbb{E}_s [\text{KL}(\pi(\cdot | s) \parallel \hat{\pi}(\cdot | s))] \right\}^{1/2} \\ &\leq C \hat{\rho}_\pi(n, \delta). \end{aligned}$$

392 Taking square roots and invoking Assumption 2, with probability at least $1 - \delta$,

$$\begin{aligned} \|\hat{u} - u^*\|_{2, \text{beh}} &\leq \sqrt{2} e^{2L} \left\{ \mathbb{E}_s [\text{KL}(\pi(\cdot | s) \parallel \hat{\pi}(\cdot | s))] \right\}^{1/2} \\ &\leq C \hat{\rho}_\pi(n, \delta). \end{aligned}$$

393 where $C := \sqrt{2} e^{2L}$. □

394 **Lemma 4** (Per-iteration error bound). *Under Assumptions 2–3, there exists a constant $C \in (0, \infty)$
395 such that, with probability at least $1 - \delta$,*

$$\|\mathcal{T}_u^\mu(\hat{Q}^{(k-1)}) - \hat{Q}^{(k)}\|_{2, \text{beh}} \leq C \hat{\rho}_\pi(n, \delta) + \hat{\rho}_Q(n, \delta) + \varepsilon_{\mathcal{F}}.$$

396 *Proof of Lemma 4.* Use $\|\cdot\|$ to denote $\|\cdot\|_{2, \text{beh}}$. Let

$$R_k := \|\mathcal{T}_u^\mu(\hat{Q}^{(k-1)}) - \hat{Q}^{(k)}\|, \quad R_k^* := \inf_{f \in \mathcal{F}} \|\mathcal{T}_u^\mu(\hat{Q}^{(k-1)}) - f\|.$$

397 By definition of the regret and Assumption 3, with probability at least $1 - \delta/2$,

$$R_k^2 - (R_k^*)^2 = \text{reg}(\hat{Q}^{(k)} | \hat{u}, \hat{Q}^{(k-1)}) \leq \hat{\rho}_Q(n, \delta/2)^2.$$

398 Since

$$R_k^2 - (R_k^*)^2 = (R_k - R_k^*)(R_k + R_k^*) \geq (R_k - R_k^*)^2,$$

399 it follows that

$$R_k \leq R_k^* + \hat{\rho}_Q(n, \delta/2).$$

400 Moreover, for any $f \in \mathcal{F}$,

$$\begin{aligned} \|\mathcal{T}_u^\mu(\hat{Q}^{(k-1)}) - f\| &\leq \|\mathcal{T}_u^\mu(\hat{Q}^{(k-1)}) - \mathcal{T}_{u^*}^\mu(\hat{Q}^{(k-1)})\| \\ &\quad + \|\mathcal{T}_{u^*}^\mu(\hat{Q}^{(k-1)}) - f\| \\ &= \|\hat{u} - u^*\| + \|\mathcal{T}_{u^*}^\mu(\hat{Q}^{(k-1)}) - f\|. \end{aligned}$$

401 Taking the infimum over $f \in \mathcal{F}$ yields

$$R_k^* \leq \|\hat{u} - u^*\| + \inf_{f \in \mathcal{F}} \|\mathcal{T}_{u^*}^\mu(\hat{Q}^{(k-1)}) - f\|.$$

402 Since ν_π is a probability measure,

$$\|h\|_{2,\text{beh}} \leq \|h\|_\infty \quad \text{for all } h,$$

403 so by the definition of $\varepsilon_{\mathcal{F}}$,

$$\inf_{f \in \mathcal{F}} \|\mathcal{T}_{u^*}^\mu(\hat{Q}^{(k-1)}) - f\| \leq \inf_{f \in \mathcal{F}} \|\mathcal{T}_{u^*}^\mu(\hat{Q}^{(k-1)}) - f\|_\infty \leq \varepsilon_{\mathcal{F}}.$$

404 Therefore, with probability at least $1 - \delta/2$,

$$\|\mathcal{T}_{\hat{u}}^\mu(\hat{Q}^{(k-1)}) - \hat{Q}^{(k)}\| \leq \hat{\rho}_Q(n, \delta/2) + \|\hat{u} - u^*\| + \varepsilon_{\mathcal{F}}.$$

405 Finally, under Assumption 2, Lemma 3, and a union bound, there exists a constant $C \in (0, \infty)$ such
406 that, with probability at least $1 - \delta/2$,

$$\|\hat{u} - u^*\|_{2,\text{beh}} \leq C \hat{\rho}_\pi(n, \delta/2).$$

407 Combining the two displays and applying another union bound gives, with probability at least $1 - \delta$,

$$\|\mathcal{T}_{\hat{u}}^\mu(\hat{Q}^{(k-1)}) - \hat{Q}^{(k)}\|_{2,\text{beh}} \leq C \hat{\rho}_\pi(n, \delta/2) + \hat{\rho}_Q(n, \delta/2) + \varepsilon_{\mathcal{F}}.$$

408 Absorbing the $\delta/2$ change into the notation proves the claim. $\square$

409 D Experiments

410 We aim to show that accurate reward recovery (up to normalization) and expert-level policies can
411 be obtained without complicated optimization, matching MaxEnt IRL when the function class is
412 correctly specified and improving upon it when generic function approximation is required.

413 We evaluate our approach in three gridworld domains with discounted demonstrations ($\gamma = 0.97$).
414 We compare (i) MaxEnt IRL with gradient-based reward fitting, and (ii) Ours, which first estimates
415 the trivial solution $u = \log \pi$ using a simple neural classifier, and then computes the normalized
416 solution by solving the corresponding fixed-point equation. For the MaxEnt baseline we implement
417 differentiable soft value iteration (temperature = 1), trained with Adam, learning-rate scheduling,
418 gradient clipping, and early stopping. Existing implementations in the `imitation` package and in
419 `qzed/irl-maxent` only support action-independent rewards and were not applicable. All methods
420 use the true transition kernel. Reward recovery is evaluated using Q-differences $Q(s, a) - Q(s, 0)$,
421 which are invariant to normalization and isolate recovery up to potential shaping (reported as RMSE
422 and correlation). Policy quality is measured by KL divergence, total variation (TV), and top-1
423 accuracy. All results are averaged over 100 reruns.

424 **Easy identifiable.** On a 4×4 torus gridworld with linear rewards, both methods achieve near-perfect
425 recovery ($\text{Corr} \geq 0.99$) and indistinguishable policies ($\text{KL} \leq 0.006$).

426 **Identifiable.** In an 8×8 gridworld with tabular linear rewards, MaxEnt IRL is competitive, but Ours
427 yields lower error (RMSE 0.0086 vs. 0.1081) with equally accurate policies.

428 **Misspecified.** With nonlinear rewards on the same 8×8 gridworld, the linear MaxEnt baseline
429 underfits (Corr 0.83, KL 0.049), while Ours with a neural policy head improves both recovery and
430 policy matching (Corr 0.73, KL 0.0032, TV 0.0083).

431 E Conclusion, extensions, and future work

432 We showed that MaxEnt IRL can be reframed as estimating a state potential that maps a trivial v -free
433 solution to one satisfying a chosen reward normalization. This yields a simple pipeline: estimate
434 the behavior policy, then run a few regression iterations to enforce the normalization and recover
435 the reward. The method is modular, uses off-the-shelf supervised learning, and admits finite-sample
436 guarantees under flexible nonparametric approximation. More broadly, our perspective highlights

Table 1: Results in three gridworld domains. Q-diff (RMSE), KL, and TV: lower is better. Corr and Top-1: higher is better. Entries are mean $\pm$ SE across reruns.

Exp.	Method	RMSE $\downarrow$	Corr $\uparrow$
Easy	MaxEnt	0.17 ± 0.03	0.996 ± 0.005
	Ours	0.13 ± 0.02	0.992 ± 0.006
Ident.	MaxEnt	0.27 ± 0.04	0.981 ± 0.005
	Ours	0.017 ± 0.003	0.999 ± 0.000
Hard	MaxEnt	0.83 ± 0.02	0.561 ± 0.031
	Ours	0.23 ± 0.01	0.980 ± 0.001

Exp.	Method	KL $\downarrow$	TV $\downarrow$	Top-1 $\uparrow$
Easy	MaxEnt	0.0057 ± 0.0007	0.045 ± 0.005	1.00
	Ours	0.0011 ± 0.0002	0.016 ± 0.002	1.00
Ident.	MaxEnt	0.0016 ± 0.0005	0.018 ± 0.004	0.86 ± 0.07
	Ours	0.0000	0.0016 ± 0.0002	0.93 ± 0.07
Hard	MaxEnt	0.029 ± 0.002	0.094 ± 0.004	0.50 ± 0.03
	Ours	0.0011 ± 0.0001	0.018 ± 0.001	0.90 ± 0.02

that behavior-policy logits already encode long-run value; IRL then attributes this value over time in a way consistent with the normalization.

We focused on Gumbel reward shocks (equivalently, entropy-regularized control). A natural extension is to allow general shock distributions with CDF F , which would modify the soft Bellman update and the trivial u^* solution while preserving the overall structure. Another direction is to replace iterative regression with a minimax estimator, analogous to minimax Q -estimators in offline RL (Uehara et al., 2021), which may relax Bellman completeness by decoupling the value and critic classes. Finally, alternative reward normalizations can be incorporated by changing the constraint used to select a representative from the potential-based equivalence class in (2).

Impact Statement.

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Abbeel, P. and Ng, A. Y. Apprenticeship learning via inverse reinforcement learning. In *ICML*, 2004.
- Agarwal, R., Schwarzer, M., Castro, P. S., Courville, A. C., and Bellemare, M. Deep reinforcement learning at the edge of the statistical precipice. *Advances in neural information processing systems*, 34:29304–29320, 2021.
- Aguirregabiria, V. and Mira, P. Dynamic discrete choice structural models: A survey. *Journal of Econometrics*, 156(1):38–67, 2010.
- Bajari, P., Hong, H., and Ryan, S. P. Identification and estimation of a discrete game of complete information. *Econometrica*, 78(5):1529–1568, 2010.
- Cao, H., Cohen, S., and Szpruch, L. Identifiability in inverse reinforcement learning. *Advances in Neural Information Processing Systems*, 34:12362–12373, 2021.
- Chen, J. and Jiang, N. Information-theoretic considerations in batch reinforcement learning. In *International conference on machine learning*, pp. 1042–1051. PMLR, 2019.

463 Ernst, D., Geurts, P., and Wehenkel, L. Tree-based batch mode reinforcement learning. *Journal of*
464 *Machine Learning Research*, 6, 2005.

465 Foster, D. J. and Syrgkanis, V. Orthogonal statistical learning. *The Annals of Statistics*, 51(3):
466 879–908, 2023.

467 Foster, D. J., Krishnamurthy, A., Simchi-Levi, D., and Xu, Y. Offline reinforcement learning:
468 Fundamental barriers for value function approximation. *arXiv preprint arXiv:2111.10919*, 2021.

469 Foster, D. J., Block, A., and Misra, D. Is behavior cloning all you need? understanding horizon
470 in imitation learning. *Advances in Neural Information Processing Systems*, 37:120602–120666,
471 2024.

472 Fu, J., Luo, K., and Levine, S. Learning robust rewards with adversarial inverse reinforcement
473 learning. In *ICLR*, 2018.

474 Fujimoto, S., Meger, D., and Precup, D. Off-policy deep reinforcement learning without exploration.
475 In *International conference on machine learning*, pp. 2052–2062. PMLR, 2019.

476 Geng, S., Nassif, H., Manzanares, C., Reppen, M., and Sircar, R. Deep pqr: Solving inverse
477 reinforcement learning using anchor actions. In *International Conference on Machine Learning*,
478 pp. 3431–3441. PMLR, 2020.

479 Haarnoja, T., Tang, H., Abbeel, P., and Levine, S. Reinforcement learning with deep energy-based
480 policies. In *International conference on machine learning*, pp. 1352–1361. PMLR, 2017.

481 Haarnoja, T., Zhou, A., Abbeel, P., and Levine, S. Soft actor-critic: Off-policy maximum entropy
482 deep reinforcement learning with a stochastic actor. In *ICML*, 2018.

483 Ho, J. and Ermon, S. Generative adversarial imitation learning. In *NeurIPS*, 2016.

484 Hotz, V. J. and Miller, R. A. Conditional choice probabilities and the estimation of dynamic models.
485 *The Review of Economic Studies*, 60(3):497–529, 1993.

486 Ionides, E. L. Truncated importance sampling. *Journal of Computational and Graphical Statistics*,
487 17(2):295–311, 2008.

488 Kallus, N. and Udell, M. Revealed preference at scale: Learning personalized preferences from
489 assortment choices. In *Proceedings of the 2016 ACM Conference on Economics and Computation*,
490 pp. 821–837, 2016.

491 Kallus, N., Puli, A. M., and Shalit, U. Removing hidden confounding by experimental grounding.
492 *Advances in neural information processing systems*, 31, 2018.

493 Kappen, H. J. Linear theory for control of nonlinear stochastic systems. *Physical Review Letters*, 95
494 (20):200201, 2005.

495 Ke, L., Choudhury, S., Barnes, M., Sun, W., Lee, G., and Srinivasa, S. Imitation learning as f-
496 divergence minimization. In *International workshop on the algorithmic foundations of robotics*,
497 pp. 313–329. Springer, 2020.

498 Koltchinskii, V. *Oracle inequalities in empirical risk minimization and sparse recovery problems:*
499 *Ecole D’Eté de Probabilités de Saint-Flour XXXVIII-2008*, volume 2033. Springer, 2011.

500 Levine, S. Reinforcement learning and control as probabilistic inference: Tutorial and review. *arXiv*
501 *preprint arXiv:1805.00909*, 2018.

502 Levine, S., Popović, Z., and Koltun, V. Nonlinear inverse reinforcement learning with gaussian
503 processes. In *NeurIPS*, 2011.

504 Mnih, V., Kavukcuoglu, K., Silver, D., Graves, A., Antonoglou, I., Wierstra, D., and Riedmiller, M.
505 Playing atari with deep reinforcement learning. *arXiv preprint arXiv:1312.5602*, 2013.

506 Mohri, M. and Rostamizadeh, A. Stability bounds for stationary φ -mixing and β -mixing processes.
507 *Journal of Machine Learning Research*, 11(2), 2010.

508 Munos, R. and Szepesvári, C. Finite-time bounds for fitted value iteration. *Journal of Machine*
509 *Learning Research*, 9(5), 2008.

510 Nachum, O., Norouzi, M., Xu, K., and Schuurmans, D. Bridging the gap between value and policy
511 based reinforcement learning. In *NeurIPS*, 2017.

512 Ng, A. Y. and Russell, S. J. Algorithms for inverse reinforcement learning. In *ICML*, 2000.

513 Ng, A. Y., Harada, D., and Russell, S. Policy invariance under reward transformations: Theory and
514 application to reward shaping. In *Icml*, volume 99, pp. 278–287. Citeseer, 1999.

515 Patterson, A., White, A., and White, M. A generalized projected bellman error for off-policy value
516 estimation in reinforcement learning. *Journal of Machine Learning Research*, 23(145):1–61, 2022.

517 Rust, J. Optimal replacement of gmc bus engines: An empirical model of harold zurcher. *Economet-*
518 *rica: Journal of the Econometric Society*, pp. 999–1033, 1987.

519 Sawant, N., Namballa, C. B., Sadagopan, N., and Nassif, H. Contextual multi-armed bandits for
520 causal marketing. *arXiv preprint arXiv:1810.01859*, 2018.

521 Skalse, J. and Abate, A. Partial identifiability and misspecification in inverse reinforcement learning.
522 *arXiv preprint arXiv:2411.15951*, 2024.

523 Skalse, J. M. V., Farrugia-Roberts, M., Russell, S., Abate, A., and Gleave, A. Invariance in policy
524 optimisation and partial identifiability in reward learning. In *International Conference on Machine*
525 *Learning*, pp. 32033–32058. PMLR, 2023.

526 Snoswell, A. J., Singh, S. P., and Ye, N. Revisiting maximum entropy inverse reinforcement learning:
527 New perspectives and algorithms. In *2020 IEEE Symposium Series on Computational Intelligence*
528 *(SSCI)*, pp. 241–249. IEEE, 2020.

529 Todorov, E. Efficient computation of optimal actions. *Proceedings of the National Academy of*
530 *Sciences*, 106(28):11478–11483, 2009.

531 Tosatto, S., Pirota, M., d’Eramo, C., and Restelli, M. Boosted fitted q-iteration. In *International*
532 *Conference on Machine Learning*, pp. 3434–3443. PMLR, 2017.

533 Tsitsiklis, J. and Van Roy, B. Analysis of temporal-difference learning with function approximation.
534 *Advances in neural information processing systems*, 9, 1996.

535 Uehara, M., Huang, J., and Jiang, N. Minimax weight and q-function learning for off-policy
536 evaluation. In *International Conference on Machine Learning*, pp. 9659–9668. PMLR, 2020.

537 Uehara, M., Imaizumi, M., Jiang, N., Kallus, N., Sun, W., and Xie, T. Finite sample analysis of
538 minimax offline reinforcement learning: Completeness, fast rates and first-order efficiency. 2021.

539 Uehara, M., Kallus, N., Lee, J. D., and Sun, W. Offline minimax soft-q-learning under realizability
540 and partial coverage. *Advances in Neural Information Processing Systems*, 36:12797–12809, 2023.

541 van de Geer, S. *Empirical Processes in M-estimation*, volume 6. Cambridge university press, 2000.

542 van der Laan, L. A researcher’s guide to empirical risk minimization. *arXiv preprint*
543 *arXiv:2602.21501*, 2026.

544 van der Laan, L. and Kallus, N. Fitted q evaluation without bellman completeness via stationary
545 weighting. *arXiv preprint arXiv:2512.23805*, 2025.

546 Wainwright, M. J. *High-dimensional statistics: A non-asymptotic viewpoint*, volume 48. Cambridge
547 university press, 2019.

548 Wulfmeier, M., Ondruska, P., and Posner, I. Maximum entropy deep inverse reinforcement learning.
549 In *AAAI*, 2016.

550 Xie, T. and Jiang, N. Q* approximation schemes for batch reinforcement learning: A theoretical
551 comparison. In *Conference on Uncertainty in Artificial Intelligence*, pp. 550–559. PMLR, 2020.

- 552 Xie, T. and Jiang, N. Batch value-function approximation with only realizability. In *International*
553 *Conference on Machine Learning*, pp. 11404–11413. PMLR, 2021.
- 554 Xie, T., Foster, D. J., Bai, Y., Jiang, N., and Kakade, S. M. The role of coverage in online reinforcement
555 learning. *arXiv preprint arXiv:2210.04157*, 2022.
- 556 Yu, B. Rates of convergence for empirical processes of stationary mixing sequences. *The Annals of*
557 *Probability*, pp. 94–116, 1994.
- 558 Zeng, S., Li, C., Garcia, A., and Hong, M. Maximum-likelihood inverse reinforcement learning with
559 finite-time guarantees. *Advances in Neural Information Processing Systems*, 35:10122–10135,
560 2022.
- 561 Zhan, W., Huang, B., Huang, A., Jiang, N., and Lee, J. Offline reinforcement learning with
562 realizability and single-policy concentrability. In *Conference on Learning Theory*, pp. 2730–2775.
563 PMLR, 2022.
- 564 Ziebart, B. D., Maas, A., Bagnell, J. A., and Dey, A. K. Maximum entropy inverse reinforcement
565 learning. In *AAAI*, 2008.
- 566 Ziebart, B. D., Bagnell, J. A., and Dey, A. K. Modeling interaction via the principle of maximum
567 causal entropy. In *ICML*, 2010.

568 A Instantiated algorithm

569 Our high-probability analysis is based on the instantiation of (2) with sample splitting and regression
570 via non-parametric least squares. This is presented in (3).

Algorithm 3 Split-Classify-Regress IRL

Inputs: transitions $\{(s_i, a_i, s'_i)\}_{i=1}^n$; reference measure $\mu(\cdot | s)$; discount factor γ ; value class $\mathcal{V}$; steps K ; a probabilistic multiclass classifier (outputs $\hat{\pi}(a | s)$). **Split:** partition the data into $\mathcal{D}^{\text{cls}}$ and $\mathcal{D}^{\text{reg}}$ (sizes $n/2$ each). **Classification on $\mathcal{D}^{\text{cls}}$:** train the classifier on $\{(s_i, a_i) \in \mathcal{D}^{\text{cls}}\}$ to obtain $\hat{\pi}(a | s)$; set $\hat{u}(s, a) = \log \hat{\pi}(a | s)$. **Folds for regression:** split $\mathcal{D}^{\text{reg}}$ into K folds $\{\mathcal{D}_k\}_{k=1}^K$ of size $\lfloor \frac{n}{2K} \rfloor$. Initialize $\hat{v}^{(0)}(s, a) = 0$. $k = 1, \dots, K$ **Regression on fold k :** $\hat{v}^{(k)} \in \arg \min_{v \in \mathcal{V}} \sum_{(s_i, a_i, s'_i) \in \mathcal{D}_k} \left\{ \underbrace{\mu[\gamma \hat{v}^{(k-1)} - \hat{u}](s'_i)}_{\text{outcome } y_i} - v(s_i, a_i) \right\}^2$. **Bellman update:** $\hat{c}_k(s) \leftarrow \mu[\gamma \hat{v}^{(k)} - \hat{u}](s)$. **Return:** $\hat{r}(s, a) = \hat{u}(s, a) + \hat{c}_K(s) - \gamma \hat{v}^{(K)}(s, a)$, $\hat{v}(s, a) = \hat{v}^{(K)}(s, a)$.

571 B Proofs

572 This appendix contains detailed proofs of all claims. Each proof is accompanied by comments on
573 intuition and scope.

574 B.1 Proof of (1)

575 We start with feasibility: with $u = \log P_{\text{data}}$, for any s' we have $\sum_{a'} \exp\{u(s', a')\} = 1$, so the RHS
576 of the soft Bellman equals 0 and $v \equiv 0$ is feasible. For optimality, write the per-sample log-likelihood

$$\ell(r, v; s, a) = r(s, a) + \gamma v(s, a) - \log \sum_{a'} e^{r(s, a') + \gamma v(s, a')}.$$

577 The right-hand side is the log of a categorical distribution $\pi(\cdot | s) \propto \exp\{r + \gamma v\}$, hence

$$\mathbb{E}[\ell(r, v; s, a)] = -\mathbb{E}_s \text{KL}(P_{\text{data}}(\cdot | s) \| \pi(\cdot | s)) + \mathbb{E}_s H(P_{\text{data}}(\cdot | s)),$$

578 maximized when $\pi(\cdot | s) = P_{\text{data}}(\cdot | s)$ for all s , which $(u, 0)$ achieves.

579 B.2 Proof of (2)

580 Let $\tilde{r} = r + c - \gamma \int c(s') P(ds' | s, a)$ and $\tilde{v} = v + \int c(s') P(ds' | s, a)$. For any s' ,

$$\sum_{a'} e^{\tilde{r}(s', a') + \gamma \tilde{v}(s', a')} = e^{c(s')} \sum_{a'} e^{r(s', a') + \gamma v(s', a')}.$$

581 Taking log and $\mathbb{E}[\cdot | s, a]$ shows the soft Bellman is preserved. For the objective, observe

$$\tilde{r}(s, a) + \gamma \tilde{v}(s, a) = r(s, a) + \gamma v(s, a) + c(s), \quad \log \sum_{a'} e^{\tilde{r} + \gamma \tilde{v}} = c(s) + \log \sum_{a'} e^{r + \gamma v},$$

582 so the per-sample log-likelihood is unchanged.

583 B.3 Proof of (1)

584 By (??), all optima are $(u, 0)$ shaped by a potential c . Enforcing the reward normalization $\int r d\mu = 0$
585 yields

$$0 = \mu u(s) + c(s) - \gamma \int \int c(s') P(ds' | s, a) \mu(da | s) = \mu u(s) + c(s) - \gamma (P_\mu c)(s),$$

586 i.e., $(I - \gamma P_\mu)c = -\mu u$. Since P_μ is a Markov operator with sup-norm ≤ 1 and $\gamma < 1$, $I - \gamma P_\mu$ is
587 invertible on bounded functions via the Neumann series, giving a unique c and hence unique (r^*, v^*) .

588 B.4 Proofs for Section 5

589 *Proof of Lemma ??.* Recall λ is stationary for the state kernel $P_\mu(s, ds') := \int P(ds' | s, a) \mu(da |$
590 $s)$, and let $\lambda \otimes \mu$ denote the induced stationary distribution on (s, a) (i.e., $a \sim \mu(\cdot | s)$ when $s \sim \lambda$).

591 By definition, v_u is the unique fixed point of $T_u : v \mapsto P\mu(\gamma v - u)$ on functions $v : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, so

$$v_u = P\mu(\gamma v_u - u) \iff (I - \gamma P\mu) v_u = -(P\mu)u.$$

592 Hence

$$v_u = -(I - \gamma P\mu)^{-1}(P\mu)u, \quad v_{u'} = -(I - \gamma P\mu)^{-1}(P\mu)u'.$$

593 Since $\gamma < 1$ and $P\mu$ is a contraction on $L^2(\lambda \otimes \mu)$ (by Jensen's inequality and stationarity of $\lambda \otimes \mu$),

$$\|(I - \gamma P\mu)^{-1}\|_{2, \lambda \otimes \mu} \leq (1 - \gamma)^{-1}.$$

594 Therefore,

$$\begin{aligned} \|v_u - v_{u'}\|_{2, \lambda \otimes \mu} &= \|(I - \gamma P\mu)^{-1}(P\mu)(u' - u)\|_{2, \lambda \otimes \mu} \\ &\leq \|(I - \gamma P\mu)^{-1}\|_{2, \lambda \otimes \mu} \|(P\mu)(u' - u)\|_{2, \lambda \otimes \mu} \\ &\leq (1 - \gamma)^{-1} \|u - u'\|_{2, \lambda \otimes \mu}, \end{aligned}$$

595 where the last inequality uses that $\|P\mu\|_{2, \lambda \otimes \mu} \leq 1$. Thus $u \mapsto v_u$ is $(1 - \gamma)^{-1}$ -Lipschitz from
596 $L^2(\lambda \otimes \mu)$ to $L^2(\lambda \otimes \mu)$. $\square$

597 *Proof of Theorem 2.* Since both (r, v) and $(u^*, 0)$ are optimal, Q_r and Q_{u^*} must induce the same
598 softmax policy. Thus,

$$\log \pi(a | s) = u^*(s, a) = Q_{u^*}(s, a) - \log \sum_{a' \in \mathcal{A}} \exp\{Q_{u^*}(s, a')\} = Q_r(s, a) - \log \sum_{a' \in \mathcal{A}} \exp\{Q_r(s, a')\}.$$

599 It follows that

$$Q_r(s, a) = Q_{u^*}(s, a) + c(s),$$

600 where we define

$$c(s) = \log \sum_{a' \in \mathcal{A}} \exp\{Q_r(s, a')\} - \Xi(r + \gamma v)(s).$$

601 Next, for a new policy π_1 , we have that $Q_{u^*}^{\pi_1}$ is the unique bounded solution to

$$Q_{u^*}^{\pi_1}(s, a) = u^*(s, a) + \gamma(\pi_1 P)Q_{u^*}^{\pi_1}(s, a).$$

602 Plugging in $u^*(s, a) = r(s, a) - c(s) + \gamma P c(s, a)$, we find that

$$Q_{u^*}^{\pi_1}(s, a) = r(s, a) - c(s) + \gamma P c(s, a) + \gamma(\pi_1 P)Q_{u^*}^{\pi_1}(s, a).$$

603 Rearranging and noting that $P c = \pi_1 P c$ since c does not depend on the action gives

$$Q_{u^*}^{\pi_1}(s, a) + c(s) = r(s, a) + \gamma(\pi_1 P)(Q_{u^*}^{\pi_1} + c).$$

604 Hence, $Q_{u^*}^{\pi_1}(s, a) + c(s)$ is a solution to the Bellman equation with reward $r(s, a)$. By uniqueness of
605 the fixed point, we conclude that

$$Q_r^{\pi_1}(s, a) = Q_{u^*}^{\pi_1}(s, a) + c(s) = Q_{u^*}^{\pi_1}(s, a) + \Xi(r + \gamma v)(s).$$

606 Thus,

$$Q_{u^*}^{\pi_1}(s, a) = Q_r^{\pi_1}(s, a) - \Xi(r + \gamma v)(s),$$

607 as desired.

608 The final result follows from the linearity of the difference in value functions with respect to the
609 difference in Q -functions. $\square$

610 *Proof of Lemma ??.* Let $e_k := \|v_{\hat{u}} - \hat{v}^{(k)}\|_{2, \lambda \otimes \mu}$. Since $v_{\hat{u}} = T_{\hat{u}} v_{\hat{u}}$ and $T_{\hat{u}}$ is a γ -contraction in v ,

$$e_k \leq \|T_{\hat{u}} v_{\hat{u}} - T_{\hat{u}} \hat{v}^{(k-1)}\|_{2, \lambda \otimes \mu} + \|\hat{v}^{(k)} - T_{\hat{u}} \hat{v}^{(k-1)}\|_{2, \lambda \otimes \mu} \leq \gamma e_{k-1} + \eta_k.$$

611 Unrolling this recursion gives

$$e_K \leq \gamma^K e_0 + \sum_{t=1}^K \gamma^{K-t} \eta_t,$$

612 which is the desired result. $\square$

613 *Proof of Theorem ??.* By the triangle inequality,

$$\|\hat{v}^{(K)} - v^*\|_{2, \lambda \otimes \mu} \leq \|\hat{v}^{(K)} - v_{\hat{u}}\|_{2, \lambda \otimes \mu} + \|v_{\hat{u}} - v^*\|_{2, \lambda \otimes \mu}.$$

614 Apply Lemma ?? to the first term and Lemma ?? with $u' = u^*$ to the second to obtain

$$\|\hat{v}^{(K)} - v^*\|_{2, \lambda \otimes \mu} \leq \gamma^K \|v_{\hat{u}} - \hat{v}^{(0)}\|_{2, \lambda \otimes \mu} + \sum_{t=1}^K \gamma^{K-t} \eta_t + \frac{1}{1-\gamma} \|\hat{u} - u^*\|_{2, \lambda \otimes \mu}.$$

615 The specialization with $\eta_t \leq \bar{\eta}$ follows by summing the geometric series. $\square$

616 *Proof of Lemma ??.* Let $u = \log \pi$, $\hat{u} = \log \hat{\pi}$, and define the likelihood ratio $\vartheta(a | s) := \hat{\pi}(a | s) / \pi(a | s)$. Since $\hat{u}, u \in \mathcal{U}$ are uniformly bounded from below and above, the log-likelihood ratio is
 617 also bounded $|\log \vartheta| \leq L$ for some $L = 2B$. The function $\phi(t) := -\log t - (t-1)$ is e^{-2L} -strongly
 618 convex on $[e^{-L}, e^L]$, hence for each s ,

$$\text{KL}(\pi(\cdot | s) \parallel \hat{\pi}(\cdot | s)) = \mathbb{E}_\pi[-\log \vartheta] = \mathbb{E}_\pi[\phi(\vartheta)] \geq \frac{e^{-2L}}{2} \mathbb{E}_\pi[(\vartheta - 1)^2].$$

620 Moreover, by the mean value theorem on $[e^{-L}, e^L]$, $|\log \vartheta| \leq e^L |\vartheta - 1|$, so

$$\mathbb{E}_\pi[(\hat{u} - u^*)^2] = \mathbb{E}_\pi[(\log \vartheta)^2] \leq e^{2L} \mathbb{E}_\pi[(\vartheta - 1)^2] \leq 2e^{4L} \text{KL}(\pi(\cdot | s) \parallel \hat{\pi}(\cdot | s)).$$

621 Averaging over s and taking square roots yields

$$\|\hat{u} - u^*\|_2 \leq \sqrt{2} e^{2L} \left(\mathbb{E}_s \text{KL}(\pi \parallel \hat{\pi}) \right)^{1/2}.$$

622 By Assumption 2 (KL generalization), $\mathbb{E}_s \text{KL}(\pi \parallel \hat{\pi}) \leq \hat{\rho}_{\mathcal{U}}(n/2, \delta)$ with prob. $\geq 1 - \delta$, hence

$$\|\hat{u} - u^*\|_2 \leq C(L) \hat{\rho}_{\mathcal{U}}\left(\frac{n}{2}, \delta\right), \quad C(L) := \sqrt{2} e^{2L}.$$

623 We turn to the second bound. As shorthand, let

$$(T_{\hat{u}} v)(s, a) := (P\mu)(\gamma v - \hat{u})(s, a) = \int [\gamma v(s', a') - \hat{u}(s', a')] \mu(da' | s') P(ds' | s, a).$$

624 By Assumption ??, $T_{\hat{u}} \hat{v}^{(k-1)} \in \mathcal{V}$ and is given by the population risk minimizer

$$\begin{aligned} T_{\hat{u}} \hat{v}^{(k-1)} &= \arg \min_{v \in \mathcal{V}} \mathbb{E}_{(s,a) \sim P_{\text{data}}} \left[(T_{\hat{u}} \hat{v}^{(k-1)})(s, a) - v(s, a) \right]^2 \\ &= \arg \min_{v \in \mathcal{V}} \mathbb{E}_{(s,a,s') \sim P_{\text{data}}} \left[\int (\gamma \hat{v}^{(k-1)}(s', a') - \hat{u}(s', a')) \mu(da' | s') - v(s, a) \right]^2. \end{aligned}$$

625 The estimator $\hat{v}^{(k)}$ is precisely the empirical risk minimizer over $v \in \mathcal{V}$ for the empirical least-squares
 626 analogue of the right-hand side. By Corollary 1 in Appendix C, for any $\delta \in (0, 1)$, with probability at
 627 least $1 - \frac{\delta}{2K}$,

$$\|\hat{v}^{(k)} - T_{\hat{u}} \hat{v}^{(k-1)}\|_2 \lesssim \rho_{\mathcal{V}}\left(m_K, \frac{\delta}{2K}\right),$$

628 where hidden constants only depend on the function-class bound B . We apply the corollary condi-
 629 tional on the training data used to form the pseudo-outcome

$$Y(s, a, s') = \int (\gamma \hat{v}^{(k-1)}(s', a') - \hat{u}(s', a')) \mu(da' | s').$$

630 By sample splitting, the data used to fit $\hat{v}^{(k)}$ are conditionally i.i.d., and the LS-ERM assumptions
 631 (bounded outcomes, convex $\mathcal{V}$) hold. Moreover, $Y(s, a, s')$ is uniformly bounded since $\hat{u}$ and $\hat{v}^{(k-1)}$
 632 are uniformly bounded by assumption on the classes $\mathcal{U}$ and $\mathcal{V}$.

633 The result now follows by collecting all bounds and noting that a union bound shows all events occur
 634 with probability at least

$$1 - \left(\frac{\delta}{2} + K \cdot \frac{\delta}{2K} \right) = 1 - \delta.$$

635 $\square$

636 *Proof of Theorem ??.* Intersect the events in Lemma ?? to obtain, with probability at least $1 - \delta$, the
 637 bounds $\nu := \|\hat{u} - u^*\|_2 \leq \hat{\rho}_U(n/2, \delta/2)$ and $\eta_k := \|\hat{v}^{(k)} - A_{\hat{u}} \hat{v}^{(k-1)}\|_2 \leq \hat{\rho}_V(m_K, \delta/(2K))$ for
 638 all k . Plug these into Theorem ?? to get

$$\|\hat{v}^{(K)} - c^*\|_2 \leq \frac{1}{1-\gamma} \hat{\rho}_U\left(\frac{n}{2}, \frac{\delta}{2}\right) + \gamma^K \|\hat{v} - \hat{v}^{(0)}\|_2 + \sum_{t=1}^K \gamma^{K-t} \hat{\rho}_V\left(m_K, \frac{\delta}{2K}\right),$$

639 and summing the geometric series yields the displayed inequality. The simplified bound for $K \simeq$
 640 $c \log n$ follows by noting that γ^K decays polynomially in n and is dominated by the statistical
 641 terms. $\square$

Proof of Lemma ??.

$$\hat{r}^{(K)} - r^* = (\hat{u} - u^*) + \mu[\gamma(\hat{v}^{(K)} - v^*) - (\hat{u} - u^*)] - \gamma(\hat{v}^{(K)} - v^*) = (\mu - I)[\gamma(\hat{v}^{(K)} - v^*) - (\hat{u} - u^*)].$$

642 Now add–subtract a Bellman step. Let

$$\varepsilon_K := \hat{v}^{(K)} - T_{\hat{u}} \hat{v}^{(K-1)}, \quad T_{\hat{u}} v := (P\mu)(\gamma v - \hat{u}).$$

643 Then

$$\hat{r}^{(K)} - r^* = (\mu - I)\left(\gamma(T_{\hat{u}} \hat{v}^{(K-1)} - v^*) - (\hat{u} - u^*)\right) + (\mu - I)[\gamma \varepsilon_K].$$

644 Thus

$$\hat{r}^{(K)} - r^* = (\mu - I)\left[-(I - \gamma P\mu)(\hat{u} - u^*) + \gamma^2(P\mu)(\hat{v}^{(K-1)} - v^*)\right] + \gamma(\mu - I)\varepsilon_K.$$

645 Assume the coverage condition

$$\kappa := \text{ess sup}_{(s,a)} \frac{dP}{d(\lambda \otimes \mu)}(s, a) < \infty$$

646 and stationarity $\lambda P\mu = \lambda$. Then, by Jensen and stationarity,

$$\|(P\mu)f\|_{2, \lambda \otimes \mu} \leq \|f\|_{2, \lambda \otimes \mu},$$

647 and by change of measure,

$$\|(P\mu)f\|_2 \leq \kappa^{1/2} \|(P\mu)f\|_{2, \lambda \otimes \mu} \leq \kappa^{1/2} \|f\|_{2, \lambda \otimes \mu}.$$

648 Hence,

$$\|\hat{r}^{(K)} - r^*\|_2 \lesssim \|\hat{u} - u^*\|_2 + \|(P\mu)(\hat{u} - u^*)\|_2 + \|(P\mu)(\hat{v}^{(K-1)} - v^*)\|_2 + \|\varepsilon_K\|_2,$$

649 and using the bounds above,

$$\|\hat{r}^{(K)} - r^*\|_2 \lesssim \|\hat{u} - u^*\|_2 + \|\hat{v}^{(K-1)} - v^*\|_{2, \lambda \otimes \mu} + \|\varepsilon_K\|_2,$$

650 with all hidden constants depending only on κ and γ .

651 $\square$

652 C High-probability bound for least squares

653 The following theorem is, up to notation, equivalent to Theorem 5.2. in [Koltchinskii \(2011\)](#)

654 **Theorem 6** (Theorem 5.2. in [Koltchinskii \(2011\)](#)). *Let $\mathcal{G}$ be a convex class of bounded functions and*
 655 *let $\hat{g}$ denote the least squares estimator of the regression function*

$$\hat{g} := \arg \min_{g \in \mathcal{G}} \frac{1}{n} \sum_{j=1}^n (Y_j - g(X_j))^2,$$

656 *where each Y_j is almost surely uniformly bounded.*

657 *Then, there exist constants $K > 0$, $C > 0$ such that for all $t > 0$,*

$$\mathbb{P}\left\{\|\hat{g} - g^*\|_2^2 \geq \inf_{g \in \mathcal{G}} \|g - g^*\|_2^2 + K\left(\hat{r}_{\mathcal{G}}(n)^2 + \frac{t}{n}\right)\right\} \leq Ce^{-t},$$

658 *where*

$$\hat{r}_{\mathcal{G}}(n) := \inf\left\{r > 0 : \hat{\mathfrak{R}}_n(\mathcal{G}, r) \lesssim r^2\right\}, \quad \hat{\mathfrak{R}}_n(\mathcal{G}, r) := \mathbb{E}_{\varepsilon} \left[\sup_{g, h \in \mathcal{G} : \|g - h\|_2 \leq r} \frac{1}{n} \sum_{i=1}^n \varepsilon_i \{g - h\}(X_i) \right].$$

659 We have the following corollary.

660 **Corollary 1** (PAC form; well-specified). *Under the conditions of Theorem 6 and assuming $g^* \in \mathcal{G}$,*
661 *for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\|\hat{g} - g^*\|_2^2 \leq K \left(\hat{r}_{\mathcal{G}}(n)^2 + \frac{1}{n} \log \frac{1}{\delta} \right).$$

662 *Proof.* By Theorem 6, for all $t > 0$,

$$\Pr \left\{ \|\hat{g} - g^*\|_2^2 \geq \inf_{g \in \mathcal{G}} \|g - g^*\|_2^2 + K \left(\hat{r}_{\mathcal{G}}(n)^2 + \frac{t}{n} \right) \right\} \leq C e^{-t}.$$

663 Under $g^* \in \mathcal{G}$, the infimum is 0. Set $t = \log(C/\delta)$ so that $C e^{-t} = \delta$. Then with probability at least
664 $1 - \delta$,

$$\|\hat{g} - g^*\|_2^2 \leq K \left(\hat{r}_{\mathcal{G}}(n)^2 + \frac{1}{n} \log \frac{C}{\delta} \right) \leq K' \left(\hat{r}_{\mathcal{G}}(n)^2 + \frac{1}{n} \log \frac{1}{\delta} \right),$$

665 absorbing $\log C$ into K' . □

666 D Use of Large Language Models

667 ChatGPT 5 was used to provide an initial skeleton of the paper, given a prompt of what the paper is
668 about and the main technical ideas. This skeleton, which had messy notation, inaccurate theorems, and
669 incorrect references, was then iterated on extensively by the human authors to achieve the submitted
670 paper. We nonetheless found the initialization very helpful to jump start the writing process.